Web design and Development Assignment (CS323)

Department of computer science

Batch DRB2302

Section A

Group members

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Submitted to: Mr.Addis Hailu

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Acknowledgment

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- Group members -

Project Documentation: Campus Connect

Problem Statement

In the modern academic environment, students are frequently confined to interactions within their own university campuses, classes, or batches. This limitation restricts opportunities for cross-campus collaboration, knowledge sharing, and networking among peers studying similar subjects or pursuing related interests. As a result, students miss out on diverse perspectives, shared resources, and potential team formations that could enhance their educational and professional growth. Campus Connect addresses this by providing a unified platform to connect students from different universities, enabling seamless communication, event organization, and community building.

Objectives

- The main goal of Campus Connect is to create an inclusive, cross-university networking platform that promotes collaboration and expands students' social and academic networks.
- To enable cross-campus connections: Allow students to discover and connect with peers from other universities based on shared interests, skills, and academic fields.
- To facilitate communication and knowledge exchange: Provide tools for messaging, group creation, and discussion to share experiences, resources, and ideas.
- To support event management: Enable users to create, promote, and participate in events that transcend individual campuses.

Functional and Non-Functional Requirements

Functional Requirements (What the system does)

- User Registration and Authentication: Students register with university details, upload ID for verification, and log in securely. Admins approve/reject registrations.
- Profile Management: Users can update profiles with bio, skills, interests, social links, and photos.
- Messaging System: One-on-one chats, group messaging, with support for text and file uploads.
- Group Management: Create, join, leave, or moderate groups tied to universities.
- Event Management: Create events, view attendance, with visibility controls
- File uploading feature
- Notifications: Real-time alerts for messages.
- Admin Controls: University admins manage approvals, bans, and block; super admins oversee everything.

Non-Functional Requirements (How the system performs)

- Scalability: Handle multiple universities and high user volumes using MongoDB's flexible schema and indexing.
- Security: Use hashed passwords, JWT for API authentication, secure file storage

- Performance: Optimize queries with indexes (e.g., on timestamps for messages/posts) to ensure fast loading of feeds and chats.
- Usability: Responsive UI, with intuitive navigation.
- Privacy: Support profile visibility settings (public, private) and data deletion options.

System Architecture (MERN Stack Overview)

Campus Connect is developed using the MERN stack, which provides a full-stack JavaScript solution for efficient development and deployment.

- MongoDB (Database): A NoSQL database that stores unstructured data in collections like users, groups, and messages. It supports complex queries and relationships via ObjectIDs, with built-in validation for data integrity.
- Express.js (Backend): Handles server-side logic, API routing, authentication middleware JWT, and integrations like file uploads using Multer.
- React (Frontend): Builds dynamic, component-based UIs for dashboards, chats, and forms, managing state with Context API for real-time updates.
- Node.js (Runtime): Powers the server, enabling asynchronous operations, REST API handling via Express.js, MongoDB/Mongoose database connections, JWT authentication, file uploads with Multer, and static asset serving for profiles and message attachments.

The architecture follows a client-server model: React frontend communicates with Express APIs, which interact with MongoDB. Security is enforced via middleware(helmet)

System Features

- Verification Workflow: Students upload IDs; status stays "pending" until admin approval, ensuring authenticity.
- Real-Time Messaging: Supports chats and group messages with file attachments, unread counts, and read receipts.
- Group Communities: Users create/manage groups for topics/projects, with roles (member, moderator, admin) and privacy settings.
- Event Organization: Create events with details like date, location, and images; users and view attendance.
- Profile Customization: Add bio, skills, interests, and social links to showcase personal branding.
- Profile Visibility: Control who can view profile (public, university, connections, private)
- File Uploads: Profile pictures, cover photos, ID verification documents, and message attachments

ER Diagram / Data Model (Optional)

The data model is relational in nature but implemented in MongoDB's document-oriented style. Key entities and relationships:

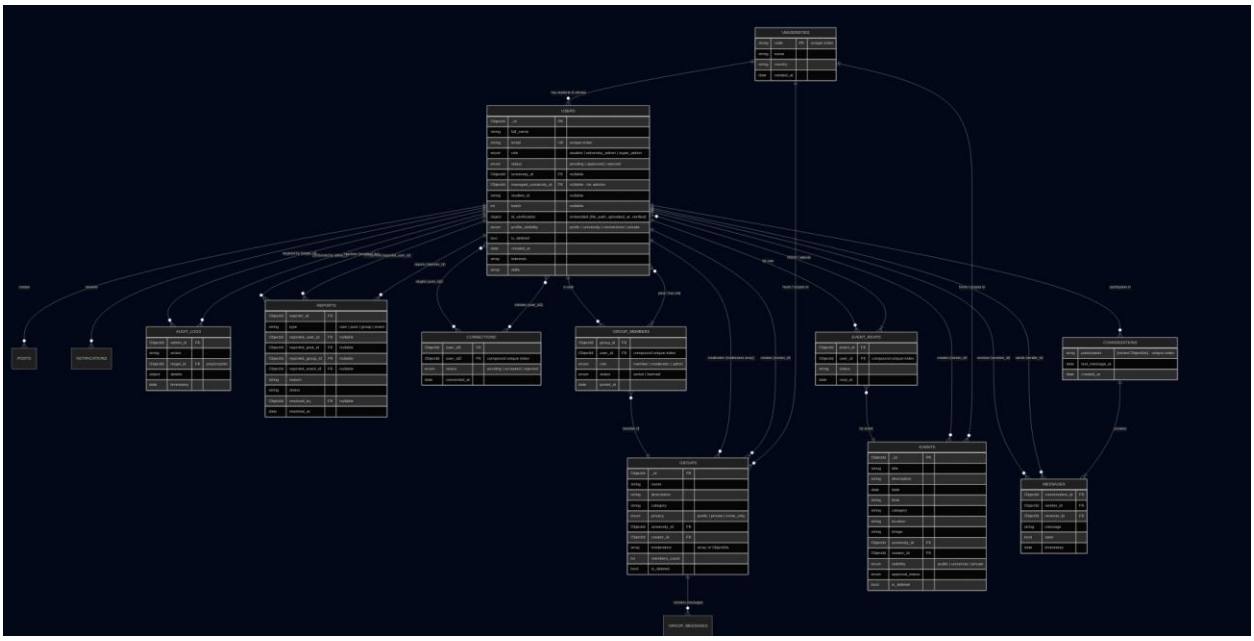
- Users (central entity): Linked to Universities (via university_id), Groups (via Group_Members), Events, Messages (as sender/receiver), Notifications (as recipient).
- Groups: Tied to Universities and Users (creator/moderators/members); contains Group_Messages.
- Events: Tied to Universities and Users (creator/attendees); has RSVPs.
- Conversations/Messages: Conversations link two Users; Messages reference Conversations.
- Universities: Parent entity for Users, Groups, and Events.

Relationships:

One-to-Many: User to Posts/Messages/Events/Groups (as creator).

Many-to-Many: Users to Groups (via Group_Members), Users to Events (via Event_RSVPs), Users to Users (via Connections).

One-to-One: User to ID_Verification (embedded object).



API Endpoints (Optional but Recommended)

The APIs are RESTful, secured with JWT authentication (auth middleware), and handle operations like file uploads.

- User Management (/api/users):GET /test, GET /me, PATCH /me, DELETE /me, GET /search, POST /me/profile-picture
- Groups (/api/groups):GET /test, GET / (list), GET /mine, POST /, DELETE /:id, POST /:id/join, POST /:id/leave, GET /:id/messages, POST /:id/messages

Additional implemented: PUT /:id/privacy, POST /:id/add-member, DELETE /:id/remove-member/:memberId

- Direct Messaging (/api/messages):GET /test, GET /conversations, GET /thread/:otherUserId, POST /send, GET /unread-count, POST /thread/:otherUserId/read, POST /contact
- Reports (/api/reports):Only GET /test is implemented ⚠ (no report submission/moderation endpoints)

- Security: All endpoints (except /test and /contact) use JWT authentication via auth.middleware.js.

- File Uploads

  Supported on:

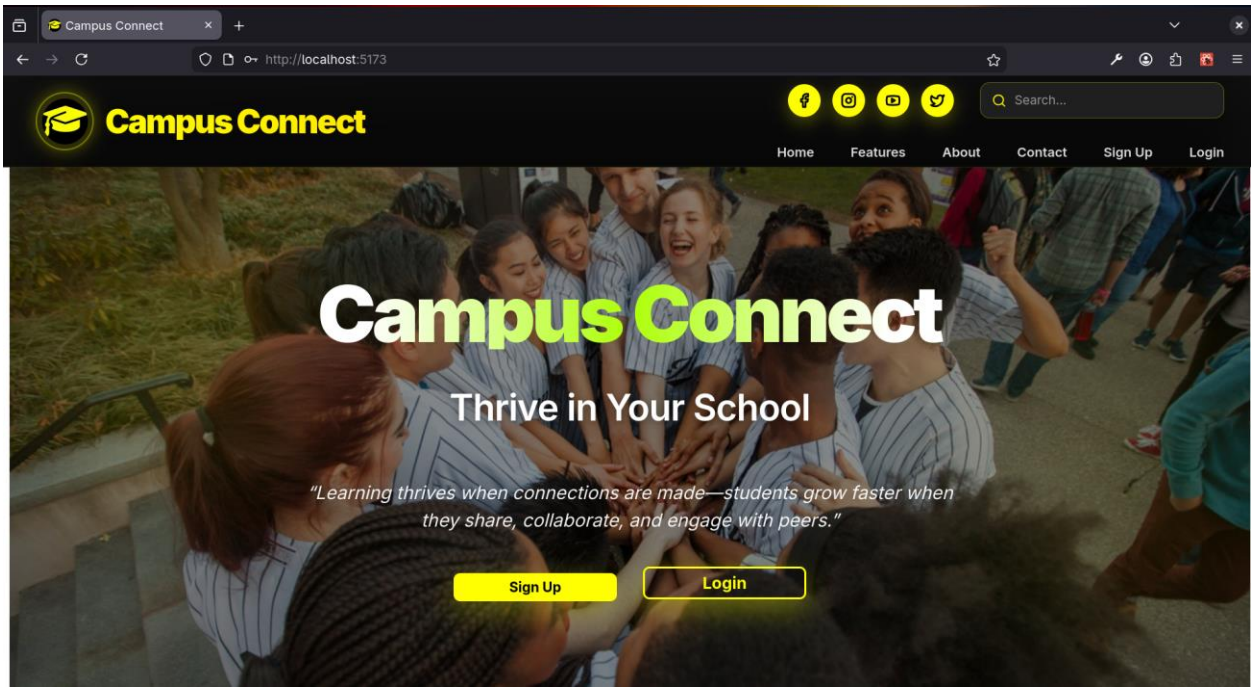
  POST /api/users/me/profile-picture (field: profile_picture)

  POST /api/groups/:id/messages (field: file)

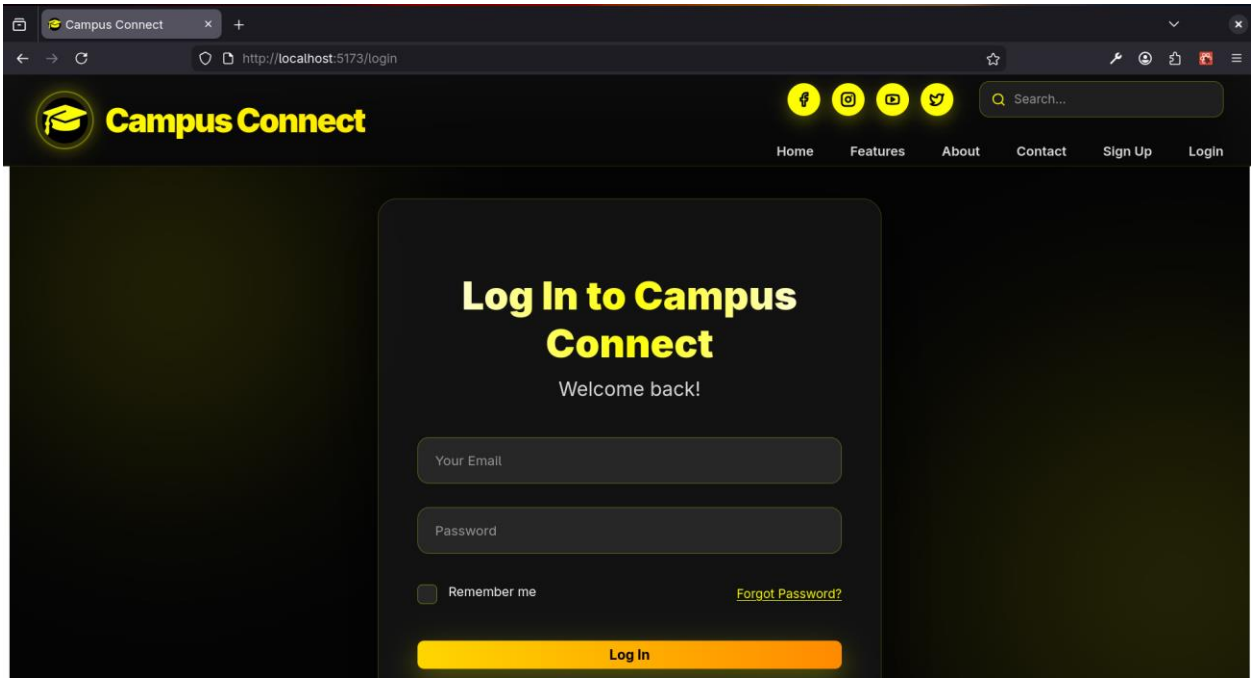
  POST /api/messages/send (field: file)

Screenshots

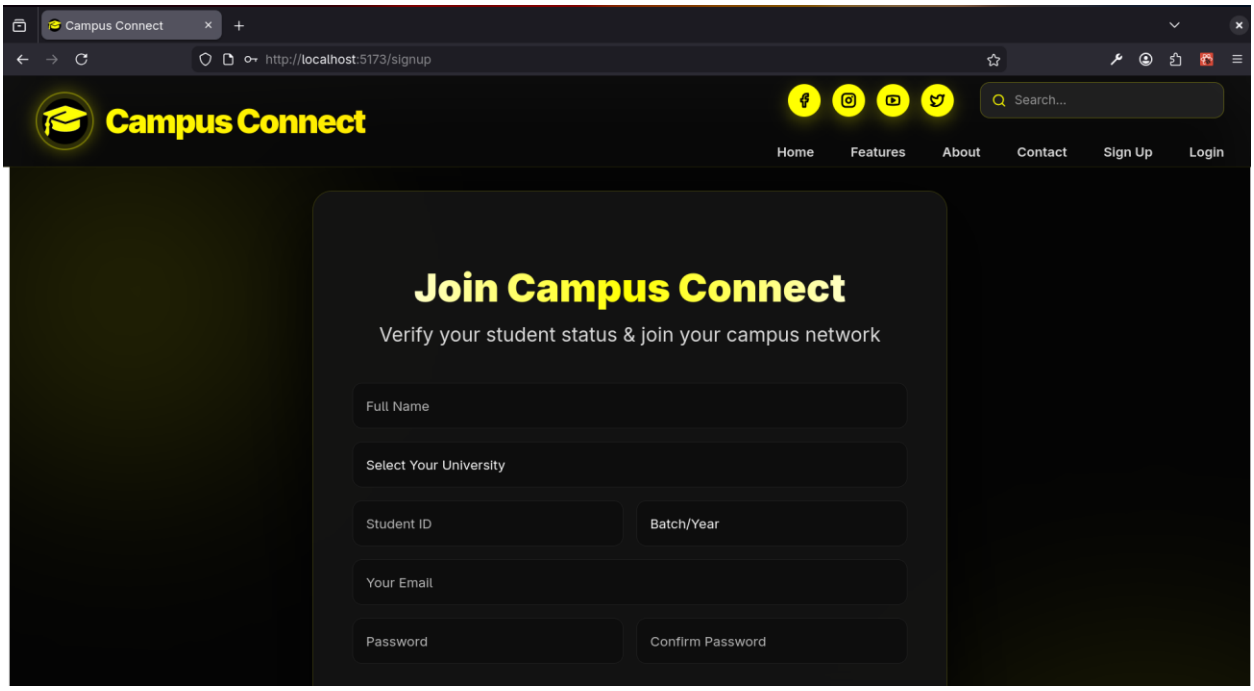
Homepage:



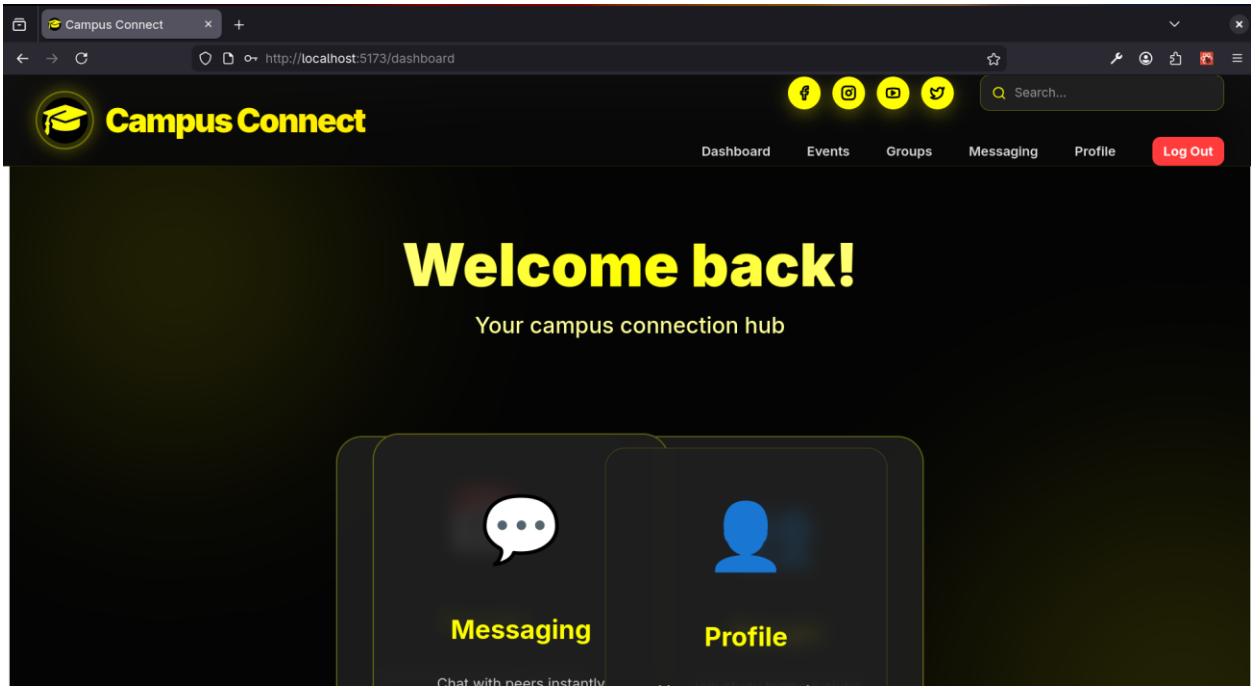
Login:



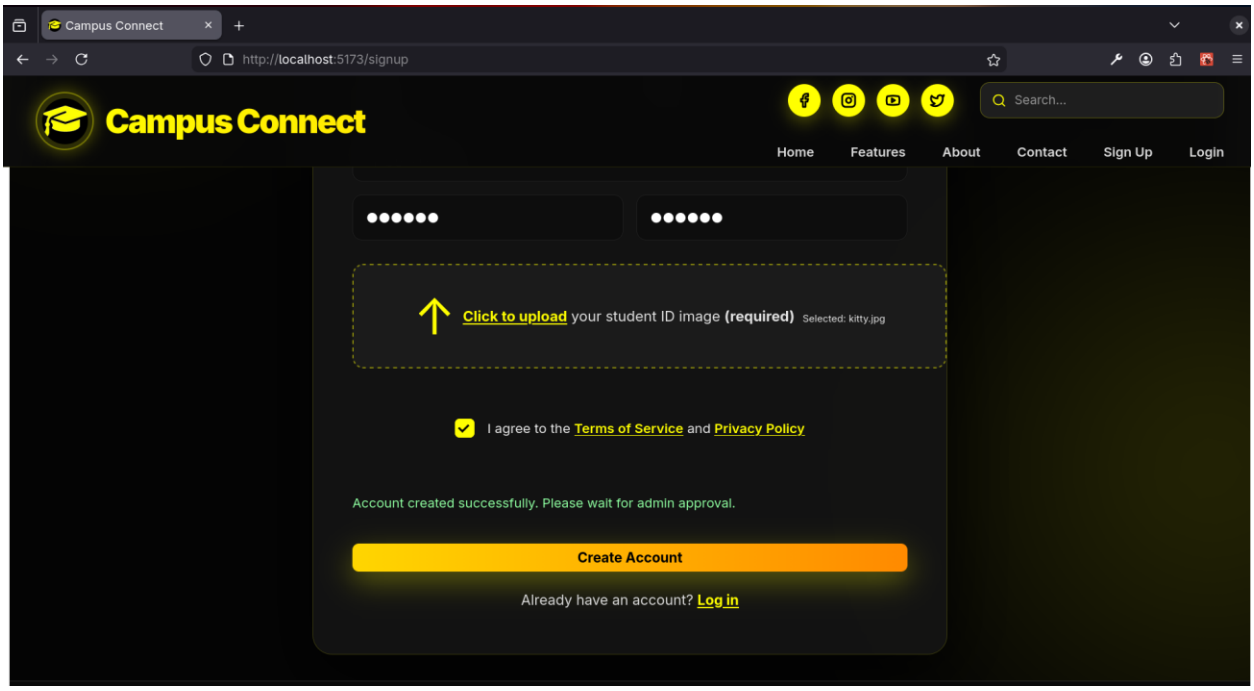
Signup:



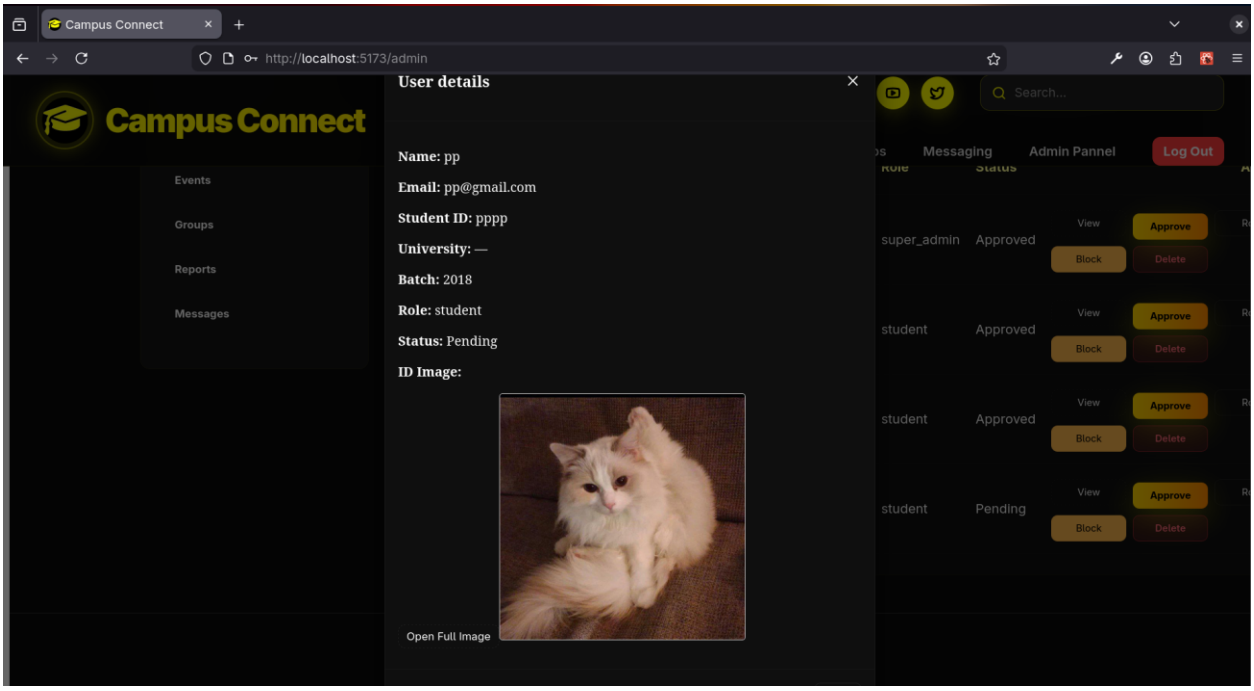
Dashboard:



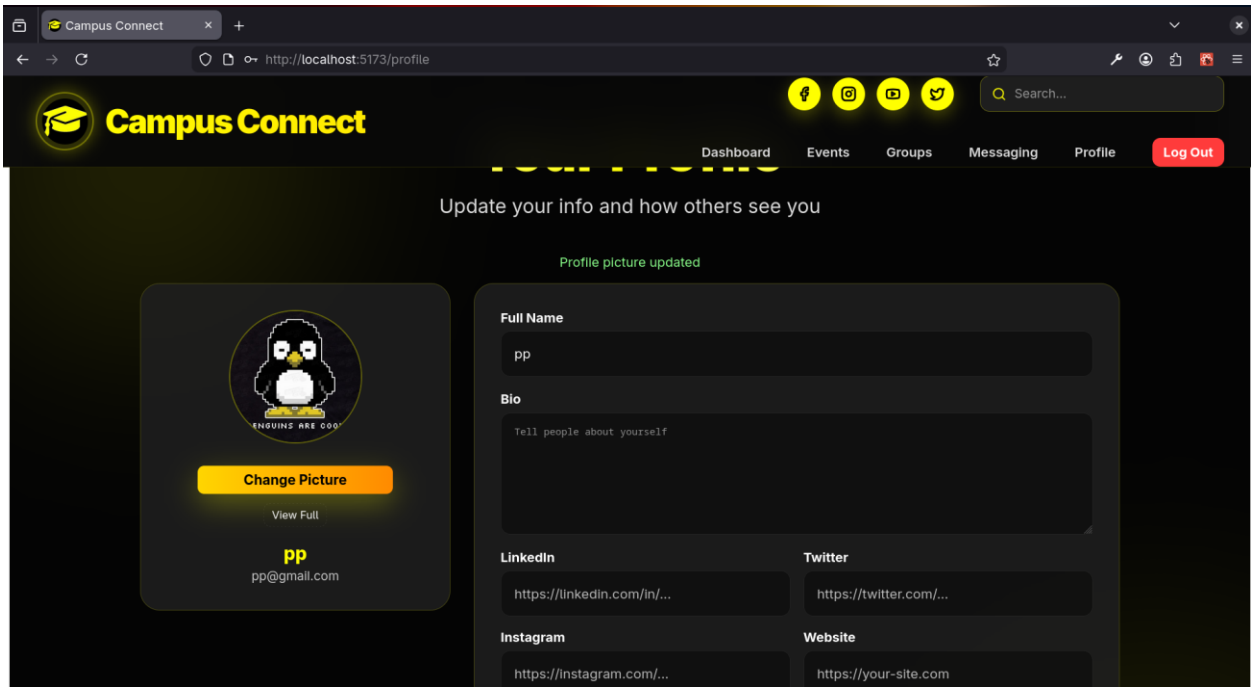
Create:



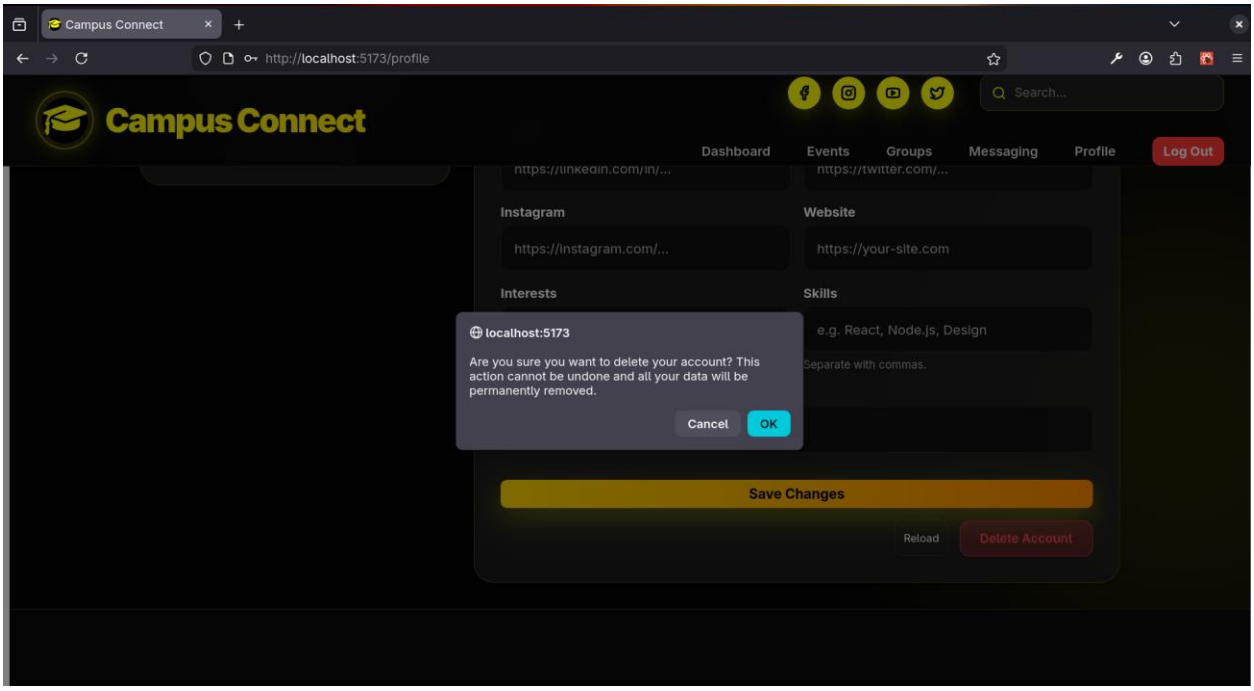
Read:


Campus Connect
http://localhost:5173/admin
Campus Connect
Events
Groups
Reports
Messages
User details
Name: pp
Email: pp@gmail.com
Student ID: pppp
University: —
Batch: 2018
Role: student
Status: Pending
ID Image:
Open Full Image
Search...
Messaging
Admin Pannel
Log Out
super_admin
Approved
student
Approved
student
Approved
student
Pending
View
Approve
Block
Delete

Update:


Campus Connect
http://localhost:5173/profile
Campus Connect
Search...
Dashboard
Events
Groups
Messaging
Profile
Log Out
Update your info and how others see you
Profile picture updated
Change Picture
View Full
pp
pp@gmail.com
Full Name
pp
Bio
Tell people about yourself
LinkedIn
https://linkedin.com/in/...
Twitter
https://twitter.com/...
Instagram
https://instagram.com/...
Website
https://your-site.com

Delete:



Conclusion & Recommendations

Conclusion

Campus Connect effectively resolves the issue of isolated student networks by offering a secure, MERN-based platform for cross-campus interaction. Through features like messaging, groups, and events, it promotes knowledge sharing and collaboration, fostering a vibrant community that enhances academic and social experiences.

Recommendations

- Mobile Expansion: Develop a mobile app using React Native for push notifications and better accessibility.
- AI Enhancements: Integrate machine learning for personalized recommendations of connections or groups based on interests/skills.

- Advanced Analytics: Add dashboards for admins to track engagement metrics.
- Integration with External Services: Connect to calendar apps (e.g., Google Calendar) for event syncing or OAuth for university logins.
- Direct School Integration: Expand partnerships with institutions to automate account creation, ensuring every enrolled student has immediate access to Campus Connect without administrative delays.